

UNITED STATES FOREST SERVICE (USFS)
INCIDENT PROCUREMENT OPERATIONS (IPO)
REQUEST FOR INFORMATION (RFI)
BATTERY PERFORMANCE

SUBMITTAL OF INFORMATION:

This RFI is for market research only and does not constitute a solicitation for proposals. The information provided may be utilized by the USFS in developing its acquisition strategy and requirements documents. It does not commit the Government to pay any cost incurred in the submission of responses or in making necessary studies or designs, nor to contract for supplies or services. No funds have been authorized, appropriated, or received for this contemplated effort. Any cost incurred shall be at the Offeror's own risk. Live demonstrations that are currently integrated with approved USFS unmanned aircraft may be requested at a future date. Responses shall include potential Offeror's Unique Entity Identifier (UEI), be in PDF format, and emailed to the Contracting Officer by 1200 hours MT, March 31, 2026, at steven.b.peterson@usda.gov.

DESCRIPTION:

The purpose of this notice is to obtain information regarding potential Offerors capable of providing innovative, high-performance battery solutions to support extended flight duration on unmanned aircraft systems (UAS). Responses may be submitted by battery manufacturers or other qualified industry partners capable of delivering flight ready or near-term battery solutions. Additionally, the Government is interested in faster charging capabilities that do not significantly degrade battery life. The focus of this RFI is on the following UAS platforms:

Platform	Manufacturer	Type	Primary Use
Alta X	Freefly Systems	III Heavy Lift	Aerial Ignition, LiDar, EO/IR, Payload adaptation

BACKGROUND:

- Alta X Demand:
 - Nominal Battery Voltage 44.4V
 - Battery Connectors XT-90
 - Required Minimum Battery Discharge Rating (Per Pack) 320 Amps per battery (assumes two batteries) 20C for a 16A-hr pack
- Current technology: 2-Tattu 16A-hr
 - Model: Tattu Plus 1.0 Compact 16000mAh 12S 15C
 - Capacity: 16000mAh (16Ah)
 - Voltage: 12S1P (44.4V)
 - Discharge Rate: 15C (30C Max Burst)
 - Dimensions: Approximately 217mm X 150mm X 80mm
 - Weight: Approximately 4250g
 - Connector: XT90
 - Balancer Connector: JST-XHR
 - Features: Smart BMS, Data Logging (CAN/SMBUS), Self-inspection, Cell Balancing

- Soliciting:
 - Technology with alternative solutions to provide increased capacity and endurance. All impacts on the current system dimensions and overall mass caused by the enhancements shall be clearly identified and corresponded.
 - United States of America domestic sources for similar make/model/performance will be prioritized. Indication of county of origin and manufacture shall be provided.

The Government's objective is to identify battery technologies with significantly improved energy density, endurance, and reliability suitable for demanding operational environments, including wildfire management and remote aerial operations.

Provide written documentation of the capability to increase battery performance and endurance. Include battery specifications. If equipment does not appear to meet minimum agency mission needs or integrations to approved USFS agency fleet, you will not be considered to perform the demonstration, should they occur. Conceptual or developmental solutions are acceptable; however, such submissions shall clearly describe the technical maturity of the solution and outline the anticipated requirements for integration with the above UAS fleet, including funding, schedule, engineering effort, and other resources required. The government is interested in battery configurations that could meet or exceed the requirements of this RFI. Any demonstration requests will require FAA-qualified Remote Pilot(s) controlling the aircraft. Proof of qualification will be required before approval of any demonstration.

TECHNICAL REQUIREMENT:

- Chemistry / Technology Type: Li-ion, Li-Po, Li-S, solid state, hybrid, etc.
- Nominal Voltage, Capacity, and Energy Density: Wh/kg, Wh/L, anticipated real-world performance-based modeling, ground testing, flight testing, or engineering estimates.
- Performance data shall be provided for operations at sea level, 2,500 ft MSL, 5,000 ft MSL, 7,500 ft MSL, and 10,000 ft MSL, with assumptions clearly stated.
- Form Factor / Packaging: Dimensions, modularity, scalability for the Alta X Aircraft
- Maximum Continuous Discharge & Peak Power Ratings: A, W, with performance over temperature range
- Battery Management System (BMS) Capabilities (if applicable): Cell balancing, State of Charge (SOC) accuracy, fault detection, fail-safe protection, and telemetry reporting to the ground control station (GCS). BMS design should support optimal energy utilization to maximize usable flight endurance while maintaining battery health and safety.
- Thermal Management Solutions: Passive/active cooling, heating requirements
- Charging system and charging time limitations

PERFORMANCE AND ENDURANCE ADVANCEMENT FROM CURRENT CONFIGURATION:

- Expected Flight Duration Improvements: Provide the anticipated percentage increase in flight duration relative to the manufacturer's currently fielded or baseline battery configuration for the applicable aircraft. Offerors shall clearly identify the baseline battery used for comparison and any assumptions applied.
- Cycle Life & Degradation: Standard cycles to 80% capacity, calendar life, etc
- Environmental Operating Envelope: Temperature, humidity, altitude, vibration tolerance

DESIGN AND OPERATION:

The design and specifications of any proposed battery system shall be compatible with the Alta X airframe and must meet aircraft center-of-gravity requirements. The installed battery system shall identify any reduction or increase in the aircraft's maximum payload capacity compared to the standard manufacturer-approved configuration. Upon Agency approval, Government officials along with the support of the submitter, will work with the aircraft manufacturer to receive "No Technical Objection" issued by the aircraft manufacturer for the battery configuration. Submittals must include the following:

- Compatibility Assessment: Alta X: hardware/electrical interfaces
- Weight, Balance & Mounting Recommendations
- Communication & Telemetry Interfaces: CAN, UART, MAVLink, etc. to GCS with aircraft % battery life displayed on GCS.
- Safety & Compliance: UN/DOT 38.3, FAA/ICAO shipping constraints, industry safety standards, etc.

Parts and Maintenance: All battery system components shall be securely installed such that no part may detach from the aircraft during operation. Components should be designed to minimize routine replacement. If required, a tool kit containing the necessary tools to perform field repairs should be provided. Recommended spare parts should be noted and readily available.

Storage: Batteries should be delivered with a case to protect the battery and any accessories while in transportation/storage. Storage instructions and maintenance of batteries while not being utilized shall be clearly identified in the maintenance manual. Solutions for effortless safe storage and maintenance compliance of batteries will be prioritized.

Emergency Operation: Offerors shall address potential fire and thermal-runaway risks associated with battery failure, including crash induced failure. Battery designs incorporating mitigation measures to reduce ignition likelihood, limit fire propagation, or enhance containment are encouraged.

Safety: Pinch points or hazards during device operation should be minimized or eliminated to prevent injury during operation or bench testing.

Support: The vendor must be able to support operators whenever the need arises. An operator's manual with pictures, diagrams, part names and identification, operating and maintenance procedures, emergency operating procedures, parts list, recommended spare parts, capacities,

weight, tool kit contents, and vendor contact information is needed. Provide the following information:

- Production Lead Times / Scalability
- Technical Support and Customization Options
- Warranty and Warranty Terms
- Past Deployments / Case Studies: UAS applications, flight test data if available

Rough Order of Magnitude (ROM): The following cost estimates may be provided. For the purposes of this RFI, a “flight cycle” is defined as the battery complement and charging capability required to support one complete aircraft sortie/mission:

- One full complement of batteries required to support a single aircraft flight cycle
- One charging system capable of supporting the aircraft’s flight cycle
- Any additional costs associated with integration of the battery system